

Quantum Transport with Kwant — companion handout

For the undergraduate course built on *Kwant — Theory and Practice*

Contents

1. What the course is	1
2. The reference card	2
3. Key equations	2
4. The models of Part II	3
5. Pitfalls (each is checked in the notebook)	3
6. Running it	3
7. Reading	4

1. What the course is

An undergraduate course on quantum transport in ten stops, taught from one executed Jupyter notebook (`Kwant_Theory_and_Practice.ipynb`, Kwant 1.5). Every figure on the slides is a numbered figure of that notebook; every equation on the slides is computed in it; the exercises at the end of each part are solved and asserted in `Kwant_Exercises_Solutions.ipynb`. The deck is a 2-D grid — sections left to right, depth top to bottom, plain language first and the mathematics last — so a lecturer can stop each section at the audience's level.

Stop	Physics	Kwant	Notebook sections
1 Lattice	Schrödinger → tight binding	Builder , lattices, hoppings	1–2
2 Leads	Landauer–Büttiker, Fisher–Lee	attach_lead , smatrix	3–4, 7
3 Shapes	resonances, billiards, parameters	shape and value functions	5
4 Spin · graphene · SC	Rashba, Dirac cones, Andreev	matrix values, honeycomb, BdG	6, 8–10
5 Magnetic fields	Peierls, Landau levels, Hofstadter	phase hoppings, wraparound	14
6 Toolbox	KPM, continuum, solvers, pitfalls	kpm , continuum , MUMPS	11–13, 15–16
7 Topology in 1D	SSH, Kitaev, Majorana wire, pump	invariants from the system	17–20
8 Topology in 2D/3D	Haldane, Kane–Mele, p+ip, BBH, Weyl, 3D TI	Berry curvature on a torus	21–26
9 Close	run it, exercises, reading	Run All	27

2. The reference card

Object	What it is	Typical line
Lattice	geometry: unit vectors + basis	<code>kwant.lattice.square(a=1, norbs=1)</code>
Site	a lattice plus integer coordinates	<code>lat(3, 4)</code>
Builder	the Hamiltonian as a mapping	<code>syst[lat(0, 0)] = 4*t;</code> <code>syst[lat.neighbors()] = -t</code>
HoppingKind	a family of bonds by displacement	<code>kwant.builder.HoppingKind((1, 0), lat)</code>
lead	a Builder with a TranslationalSymmetry	<code>syst.attach_lead(lead)</code>
finalized system	immutable, solvable	<code>fsyst = syst.finalized()</code>
scattering matrix	S at one energy	<code>sm = kwant.smatrix(fsyst, E, params=...)</code>
conductance	transmission $0 \rightarrow 1$	<code>sm.transmission(1, 0)</code> (units of e^2/h)
wavefunction	scattering states from lead 0	<code>kwant.wave_function(fsyst, E)(0)</code>
observables	density / current operators	<code>kwant.operator.Density(fsyst,)</code>
bands	lead dispersion	<code>kwant.physics.Bands(fsyst.leads[0])</code>
spectral density	KPM	<code>kwant.kpm.SpectralDensity(fsyst)</code>
continuum model	symbolic $\rightarrow$ lattice	<code>kwant.continuum.discretize(hamiltonian, grid=a)</code>
periodic system	symmetry $\rightarrow$ momentum parameter	<code>kwant.wraparound.wraparound(syst)</code>

Value rule: a value on a site or bond is a number, a matrix (spin, BdG), or a function whose first argument(s) are the site(s) and whose other named arguments are passed at solve time through `params=`.

3. Key equations

Discretisation (2D, spacing a). Hopping $t = \hbar^2/(2ma^2)$, on-site $\varepsilon_i = 4t + V(\mathbf{r}_i)$; $E(\mathbf{k}) = 4t - 2t(\cos k_x a + \cos k_y a) \approx \hbar^2 k^2/2m$.

Landauer–Büttiker. $G = \frac{e^2}{h} \sum_n T_n = \frac{e^2}{h} \text{Tr}(t^\dagger t)$; unitarity $S^\dagger S = 1$; sum rule $\sum_n (T_n + R_n) = N$.

Fisher–Lee. $S = -1 + i \Gamma^{1/2} G^r(E) \Gamma^{1/2}$, with $G^r = (E - H - \Sigma)^{-1}$, $\Sigma = \sum_{\text{leads}} V^\dagger g V$, $\Gamma = i(\Sigma - \Sigma^\dagger)$.

Bogoliubov–de Gennes. $H_{\text{BdG}} = \begin{pmatrix} H - \mu & \Delta \\ \Delta^* & -(H - \mu)^* \end{pmatrix}$, $P H_{\text{BdG}} P^{-1} = -H_{\text{BdG}}$; sub-gap conductance $G = \frac{e^2}{h} [N - R_{ee} + R_{he}]$, perfect Andreev reflection $G = 2Ne^2/h$.

Peierls substitution. $t_{ij} \rightarrow t_{ij} \exp(\frac{ie}{\hbar} \int_j^i \mathbf{A} \cdot d\mathbf{l})$; the product of the phases around a plaquette is $e^{2\pi i \Phi/\Phi_0}$, $\Phi_0 = h/e$. Harper: $-t[\psi_{y+1} + \psi_{y-1}] - 2t \cos(k_x a - 2\pi\alpha y) \psi_y = E \psi_y$.

KPM. $\rho(E) = \frac{1}{\pi \sqrt{1 - E^2}} \left[g_0 \mu_0 + 2 \sum_{n \geq 1} g_n \mu_n T_n(E) \right]$, $\mu_n = \text{Tr} T_n(H)$ (stochastic trace, Jackson kernel g_n).

SSH. $H(k) = \mathbf{d}(k) \cdot \boldsymbol{\sigma}$, $\mathbf{d} = (t_1 + t_2 \cos k, t_2 \sin k, 0)$; winding $\nu = \frac{1}{2\pi i} \oint dk \partial_k \log(d_x + i d_y)$; Zak phase $\gamma = \pi \nu$; zero modes at the ends iff $t_1 < t_2$.

Kitaev. $c_j = (\gamma_{j,1} + i\gamma_{j,2})/2$; $(-1)^\nu = \text{sgn Pf}[A(0)] \cdot \text{sgn Pf}[A(\pi)]$; topological iff $|\mu| < 2t$. Majorana nanowire: gap closes at $B_c = \sqrt{\Delta^2 + \mu^2}$; zero-bias peak $2e^2/h$.

Chern number (Fukui–Hatsugai–Suzuki). $U_i(\mathbf{k}) = \langle u_{\mathbf{k}} | u_{\mathbf{k}+\delta_i} \rangle / |\cdot|$, $F(\mathbf{k}) = \arg [U_x(\mathbf{k}) U_y(\mathbf{k} + \delta_x) U_x(\mathbf{k} + \delta_y)^{-1} U_y(\mathbf{k})^{-1}]$, $C = \frac{1}{2\pi} \sum_{\mathbf{k}} F(\mathbf{k}) \in \mathbb{Z}$; $\sigma_{xy} = (e^2/h) \sum_{\text{occ}} C_n$.

Fu–Kane. $(-1)^{\nu_0} = \prod_{\text{TRIM}} \prod_{\text{occ}} \xi$ (parity eigenvalues at the time-reversal-invariant momenta).

4. The models of Part II

Model	Lattice	Invariant	Boundary state	Phase condition (notebook)
SSH	chain, 2 sites/cell	winding number (chiral)	end zero modes	$t_1 < t_2$
Kitaev chain	chain, p-wave BdG	(Pfaffian)	Majorana end modes, ZBP $2e^2/h$	$ \mu < 2t$
Majorana nanowire	chain, Rashba + Zeeman + s-wave		Majorana end modes	$B > \sqrt{\Delta^2 + \mu^2}$
Rice–Mele pump	chain in (k, t)	Chern number	edge-state ladder, one charge per cycle	cycle encloses (0, 0)
Haldane	honeycomb, complex NNN	Chern number	chiral edge mode, $G = e^2/h$	$ M < 3\sqrt{3} t_2 \sin \phi$
Kane–Mele	honeycomb, spinful	(spin Chern mod 2)	helical edge pair, $G = 2e^2/h$	$ M < M_c$
p+ip	square, spinless BdG	BdG Chern	chiral Majorana edge	$\mu \in (-4t, 0) \cup (0, 4t)$
BBH quadrupole	square, 4 sites/cell, flux	nested Wilson loop, $q = \frac{1}{2}$	four corner zero modes	$\gamma < \lambda$
Weyl semimetal	cubic, 2 bands	sliced Chern numbers	Fermi arcs	between the nodes
3D TI	cubic, 4 bands	Fu–Kane	single surface Dirac cone, $G = 2e^2/h$	$\nu_0 = 1$ ($m = 2$)

5. Pitfalls (each is checked in the notebook)

- **MUMPS is not re-entrant:** never call `kwant.smatrix` from threads with MUMPS installed — the kernel dies without a traceback. Use processes, a serial loop, or `kwant.solvers.sparse` (SuperLU) in threads.
- **numpy 2.5** breaks `kwant.physics.magnetic_gauge` in Kwant 1.5.0; pin `numpy<2.5` (the installer does) or avoid the helper.
- **Energies inside a gap:** `num_propagating` is zero, `S` is empty.
- **Degenerate BdG pairs** come with run-dependent signs; compare $|E|$.
- **site_color** callables receive a `Site` for a Builder but an index for a finalized system.
- **Always assert an identity:** unitarity, the sum rule, particle–hole symmetry, current conservation, an integer invariant.

6. Running it

Windows: `.\install_kwant_windows.ps1`, then `python verify_kwant.py` (ends “physically correct”). Elsewhere: `conda create -n kwant -c conda-forge kwant "numpy<2.5" scipy matplotlib sympy python-mumps ipykernel jupyterlab`, then register the kernel `kwant`. Open the notebook with the kernel *Python (kwant)* and **Run All** (about 4½ minutes with MUMPS; the solutions notebook about 2). `python -m pytest tests` checks counts, licence texts and documentation; `python course/build_course.py` rebuilds this handout, the deck and the lecturer notes from the executed notebook.

7. Reading

Groth, Wimmer, Akhmerov, Waintal, *New J. Phys.* **16**, 063065 (2014) — Kwant. Datta, *Electronic Transport in Mesoscopic Systems* (1995). Nazarov & Blanter, *Quantum Transport* (2009). Asbóth, Oroszlány, Pályi, *A Short Course on Topological Insulators* (Springer 2016; arXiv:1509.02295). topocondmat.org (TU Delft). Bernevig & Hughes, *Topological Insulators and Topological Superconductors* (2013). The notebook's section 27 lists the paper behind every model.

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